

Opeyemi Peter Ojajuni, PhD

Data Scientist | Forecasting, Experimentation & Applied Machine Learning

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SUMMARY

Data scientist (PhD) with industry experience at Amazon, working across forecasting, ML modeling and experiment evaluation. At Amazon Science, wrangled 1M+ logistics and transactional records with Python and SQL and built capacity-forecasting models with quantified uncertainty for capital-investment planning. Evaluated a VR instruction intervention across 987 users over four years (quasi-experimental, ASEE 2026). Builds end-to-end prototypes, from a gradient-boosted cost-to-serve model ($R^2=0.99$ on synthetic data) to a citation-grounded RAG assistant. AWS Certified AI Practitioner.

TECHNICAL SKILLS

Languages & data: Python, SQL, PySpark (learning projects), Git/GitHub

Machine learning: scikit-learn, gradient boosting, Random Forest, XGBoost, SVM, logistic regression, CNN

Forecasting & causal inference: time series (ARIMA), capacity and demand forecasting, confounder-adjusted regression, quasi-experimental design

Statistics: factor analysis, chi-square, effect sizes, pre/post program evaluation

GenAI: RAG, LangChain, FAISS, OpenAI API

Visualization & apps: Tableau, Power BI, Streamlit

EXPERIENCE

Postdoctoral Research Manager, SU XR Jags Research Laboratory

Feb 2024 – Present

Southern University and A&M College, Baton Rouge, LA

- Evaluated a VR/CAVE-based engineering-instruction intervention against traditional instruction for 987 students over four years (Fall 2021 – Spring 2025) using a quasi-experimental, between-subjects design (ASEE 2026).
- Evaluated a VR professional-development program (NTIA Connecting Minority Communities) for 88 teachers across two 2025 cohorts (pre/post surveys, 20-item self-efficacy battery, open responses; Kirkpatrick model); Fall cohort self-efficacy rose from 5.39 to 8.72 on a 1–10 scale (+62%), and 85% of Spring participants planned to implement VR that school year (self-reported, no control group).
- Built a mobility-aware federated intrusion-detection pipeline for connected vehicles (VeReMi data), benchmarked against FedAvg, FedProx, FedNova and FedBuff; reached 0.92 malicious recall and ROC AUC 0.79 on a frozen 199,663-sample test set with ~53% estimated communication savings, and documented the high false-positive rate as the blocker to deployment.
- Co-developed a reality-capture → digital-twin → VR → 3D-print learning workflow; in a 68-student quasi-experiment, pass rates rose from 63.6% to 87.0% ($\chi^2(5) = 13.98$, $p = .016$) (journal manuscript, 2026).
- Built, trained and validated deep-learning (CNN) and classical ML models (SVM, XGBoost, Random Forest, KNN) on multimodal data, including CNN work for the Louisiana Department of Health ICLERT project.
- Lead writer on 9 grant proposals (NASA, ONR, NSF, USDA, NNSA); largest request \$1M.

Data Scientist Intern, Amazon (Amazon Science – Mechatronics & Sustainable Packaging)

Jun 2023 – Sep 2023

Seattle, WA

- Wrangled 1M+ logistics and transactional records with Python and SQL; built capacity-forecasting models with quantified uncertainty to inform capital-investment planning.
- Ran the end-to-end analytics workflow for an automated-fulfillment initiative: problem framing, hypothesis design, modeling, validation, and leadership-facing analysis.
- Built data-transformation frameworks and simulation-ready analytical layers that made models more robust to operational variability and let product stakeholders self-serve the analysis.

Graduate Research Assistant, Southern University and A&M College

Aug 2018 – Dec 2023

Baton Rouge, LA

- Built a Random Forest student-success classifier on the UCI Student Performance data and shipped it as a public Streamlit web app (SITE 2023).
- Co-authored an ARIMA-based analysis of minority graduate trends in construction-management programs (ASEE 2023).
- Validated a 29-item, 5-factor computational-thinking scale with engineering students using factor analysis (ASEE 2024), then measured medium effect sizes for algorithmic thinking and problem-solving under CAVE instruction (ASEE 2025).

Earlier: Network & Security Engineer, Staunton and Lycett Ltd, Nigeria (Mar 2015 – Feb 2018)

SELECTED PROJECTS

- Built an account-level cost-to-serve prediction pipeline (scikit-learn, gradient boosting vs. linear regression baseline) for a route-based service business, achieving $R^2=0.99$ vs. 0.82 for the baseline by modeling nonlinear cost interactions in route density, disposal distance, and facility ownership (synthetic data).
- Built a confounder-adjusted logistic regression model to estimate segment-specific price elasticity from observational data, correcting a naive estimate that was wrong-signed (-6.19) due to treatment-assignment confounding, and recovering correctly-ordered, validated elasticity coefficients for two customer segments.
- Built an airline revenue-management model (booking curves, demand forecasting, overbooking policy) showing ~10% revenue lift in simulation on synthetic data.
- Built InsightBot, a retrieval-augmented research assistant (LangChain, OpenAI embeddings, FAISS, Streamlit) that ingests article URLs and answers with source citations; public demo.
- Built an interactive HIV/AIDS dashboard covering 150+ countries (2000–2024) with UNAIDS 95-95-95 target tracking.

EDUCATION

PhD, Science & Mathematics Education, Southern University and A&M College, 2023. Dissertation: Exploring the Impact of Immersive Technology on Engineering Students' Computational Thinking Skills

MBA (in progress), Southern University and A&M College

MSc Mobile & Satellite Communication, University of Surrey, UK, 2014 · BEng Computer Engineering, Covenant University, Nigeria, 2012

CERTIFICATIONS

AWS Certified AI Practitioner (active through Jul 2028) · IBM Artificial Intelligence Practitioner (2022)

SELECTED PUBLICATIONS (20 PUBLICATIONS, 125 CITATIONS)

Ojajuni, O. P., Lawson, A. H. (2026). NSF IUSE: Evaluating academic performance in VR-enhanced engineering education: a quasi-experimental study. ASEE 2026.

Ojajuni, O., Ayeni, F., Akodu, O., et al. (2021). Predicting student academic performance using machine learning. ICCSA 2021, 481–491.